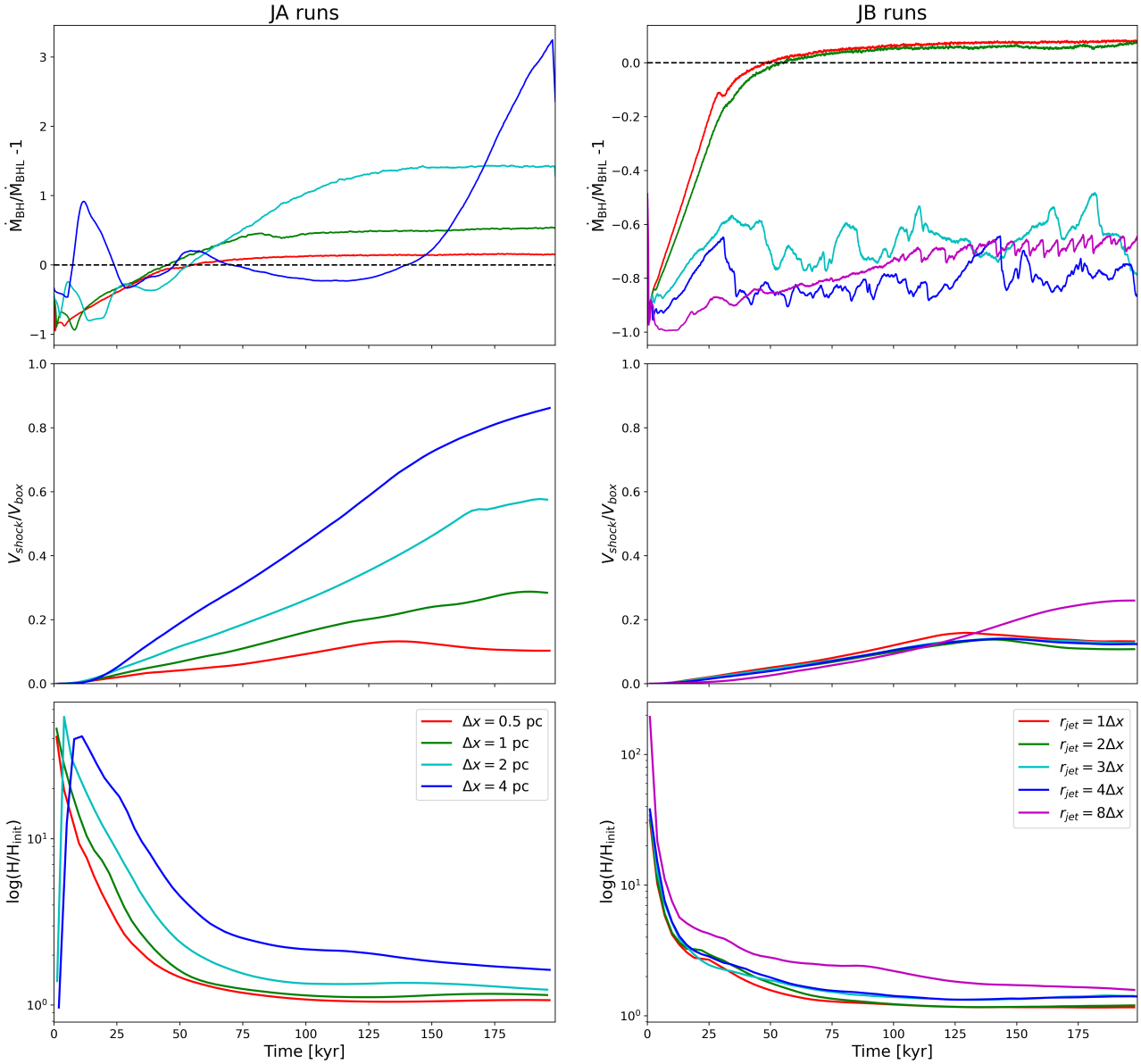

Figure 14. Evolution of accretion rate, shocked volume and enthalpy for the JA and JB runs.

the jet power for injection. Figure 19 compares the effect of the different approaches on the evolution of the accretion rates.

Using the control surface method produces a relatively steady accretion rate after an initial ramping up, while using the Bondi approach causes the accretion rate to initially drop before recovering to a value of about twice the expected rate. This suggests that the Bondi approach overestimates the initial accretion rate and thus the amount of feedback, blowing material away from the black hole and dropping the accretion rate temporarily. The control surface method only accounts for the fraction of solid angle not occupied by outflowing gas, so it allows the accretion rate to ramp up slowly even in the presence of a jet.

It is surprising that the control surface method reproduces the expected Bondi accretion rate while the Bondi method itself overestimates the accretion rate. The Bondi accretion solution does not account for outflow. Since outflow exists in the simulation, assuming that accretion rate must therefore overestimate the true accretion rate unless the outflow lowers the density or increases the sound speed within the averaging region. The control surface method permits the inflow in directions away from the jet to adjust itself in a way that somehow reproduces the Bondi rate. We would argue

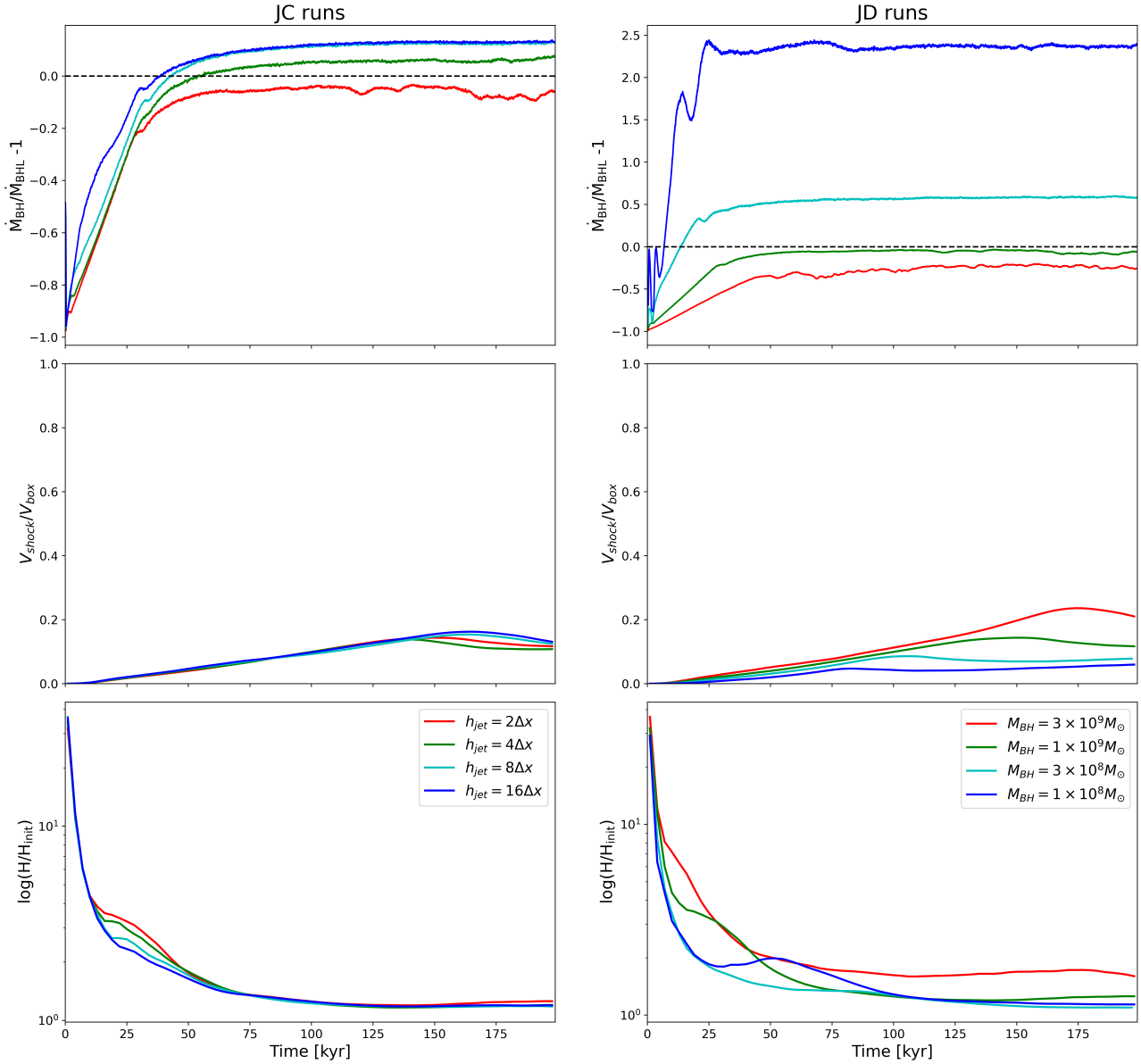

Figure 15. Evolution of accretion rate, shocked volume and enthalpy for the JC and JD runs.

that the latter behavior is more likely to be physically correct, as it relies more on the resolved non-jet flows outside the control surface than on the barely resolved resetting region.

The accretion behavior affects the structure of the jet, as seen in Figure 20. Since the average accretion rate is larger for the Bondi case, the feedback energy deposited is also greater, so at a given late time in the run the head of the jet becomes wider and the central channel becomes more prominent.

Our results are consistent with those of Negri & Volonteri (2017), who concluded that with the Bondi formalism in high resolution simulations, the BH accreted mass is generally overestimated. On the other hand, underresolved simulations underestimate the accreted mass; this is the reason for using a boost parameter. Due to the complex interplay between accretion and feedback, there is no reason to expect that the Bondi formula accurately predicts the accretion rate in realistic situations.

4. DISCUSSION AND COMPARISON WITH PREVIOUS WORK

4.1. *Comparison with Smaller-Scale Jet Simulations*

To measure the convergence of our jet model, we compare its stability, collimation, and mass entrainment

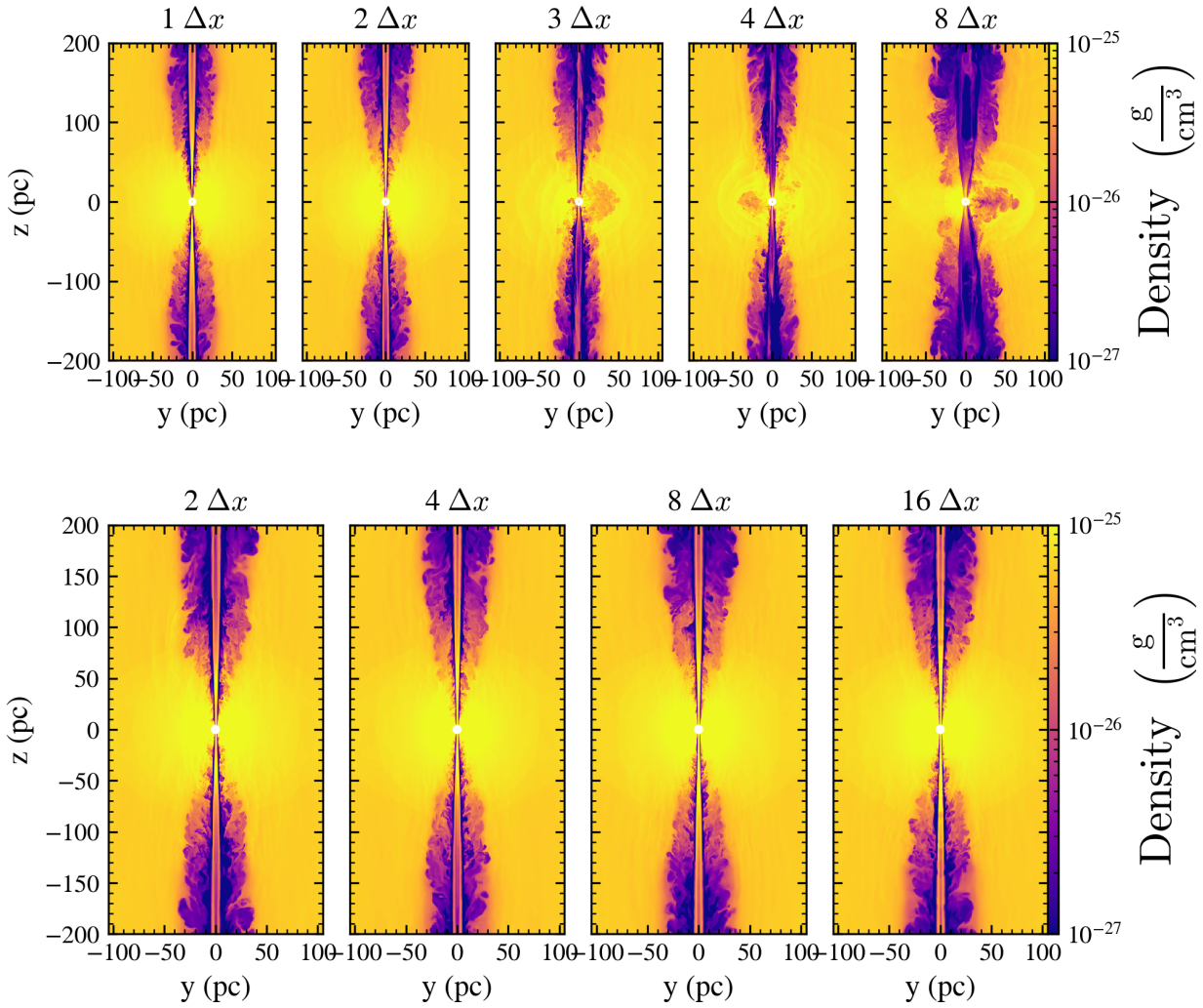

Figure 16. Gas density in the yz plane at $t = 0.2$ Myr for runs with fixed control surface size and resolution but varying radius and height of the jet injection cylinder. Top panel shows runs varying $r_{\rm jet}$ (JB runs) and bottom panel shows runs varying $h_{\rm jet}$ (JC runs).

properties with jets we observe in parsec- and smaller-scale simulations.

The comparison with GRMHD simulations can be difficult, since these simulations usually do not account for any cooling, radiative processes, or interaction between jets and the surrounding medium. However, we can aim to build a qualitative picture of how accretion powers the launching of the jets. In GRMHD simulations, usually the accretion disks start at a low accretion rate as the magnetorotational instability builds up gradually inside the torus, and after a certain time scale (around $10^4\ r_g/c$, where $r_g \equiv GM/c^2$) the accretion rate steadily grows until it saturates, after which the accretion rate remains steady. This behavior is captured by our accretion model, where under the control surface criterion the accreted material takes time to build up enough energy to launch the jets. However, the steady state value for accretion rate in GRMHD simulations (e.g. Tchekhovskoy 2015) (usually rest-mass accretion rate) tends to be 10-100 times the Bondi accretion rate, due to the scale difference.

As the jet propagates, we are also able to observe some signatures similar to MHD jets. In general, 3D relativistic, magnetized jet flows can exhibit many different kinds of instabilities (for analysis of jet instabilities, see Chow et al. 2023). Instabilities such as current-driven instability and rotation-induced instability mainly occur in magnetically dominated jets when the current flow is high, which are beyond the scope of this work. However, the Kelvin-Helmholtz instability (KHI), which occurs at the contact discontinuity or the relatively-flowed shear layers, does not depend strongly on magnetic properties